Technical Patterns and News Sentiment in Stock Markets*

Markus Leippold† Qian Wang‡ Min Yang§

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Abstract

This paper explores the effectiveness of technical patterns in predicting asset prices and market movements, emphasizing the role of news sentiment. We employ an image recognition method to detect technical patterns in price images and assess whether this approach provides more information than traditional rule-based methods. Our findings indicate that many model-based patterns yield significant returns in the US market, whereas top-type patterns are less effective in the Chinese market. The model demonstrates high accuracy in training samples and strong out-of-sample performance. Our empirical analysis concludes that technical patterns remain effective in recent stock markets when combined with news sentiment, offering a profitable portfolio strategy. Moreover, we find patterns better predict returns for firms with high momentum, institutional ownership, and prior patterns in US, while in China, they are more effective for small firms with high momentum and institutional ownership. This study highlights the potential of image recognition methods in market data analysis and underscores the importance of sentiment in technical analysis.

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†University of Zurich, Department of Finance, and Swiss Finance Institute (SFI), Plattenstrasse 14, 8032 Zurich, Switzerland; markus.leippold@df.uzh.ch.

‡University of Zurich, and Inovest Partners AG. qian.wang@uzh.ch

§University of Zurich, Department of Finance, and Swiss Finance Institute (SFI), Plattenstrasse 14, 8032 Zurich, Switzerland; min.yang2@df.uzh.ch.

1 Introduction

The Efficient Market Hypothesis (EMH) posits that asset prices reflect all available information, implying that past price movements cannot be used to predict future returns (Malkiel, 1989). This fundamental principle in financial economics casts doubt on the efficacy of technical analysis, a practice that relies on identifying recurring patterns and trends in historical price data to forecast future market movements. If markets are truly efficient, as the EMH suggests, then technical analysis should be incapable of generating consistent excess returns.

Despite these theoretical arguments, technical analysis remains a widely used tool among practitioners, and many traders and investors rely on technical indicators and patterns to inform their investment decisions (see, e.g., Taylor and Allen, 1992, Gehrig and Menkhoff, 2006). The persistent popularity of technical analysis suggests that market participants may perceive value in these patterns, even if their predictive power cannot be fully reconciled with the strict assumptions of the EMH. Moreover, the effectiveness of technical analysis may vary depending on firm characteristics even though it is not profitable across all stocks as indicated by some literature supporting EMH.

Furthermore, empirical studies have provided compelling evidence that certain technical strategies can indeed achieve significant out-of-sample returns, challenging the notion of absolute market efficiency (Neely et al., 1997, Hsu et al., 2010, Huang and Huang, 2020, Han et al., 2021, Huang et al., 2019). These findings suggest that technical patterns might capture subtle market dynamics and investor behavioral biases that are not fully accounted for by traditional fundamental analysis or efficient market models.

One possible explanation for the apparent efficacy of technical analysis lies in the realm of behavioral finance. This field explores the psychological factors that influence investor behavior and how these factors can lead to deviations from rational decision making. Technical patterns, often rooted in the collective psychology of market participants, may reflect these behavioral biases, providing clues about shifts in investor sentiment and potential turning points in the market (Peterson, 1999).

This paper examines the relevance of technical *reversal patterns*, as defined by Lo et al. (2000), and argues for a novel approach that combines the power of image recognition with

news sentiment analysis to enhance their identification and potential profitability. We contend that traditional rule-based methods for recognizing these patterns, while offering a degree of objectivity, can be inherently limited by the noise and parameter sensitivity inherent in financial data. These methods often rely on specific thresholds and criteria that can be difficult to calibrate consistently across different markets and time periods.

In addition, traditional rule-based methods may struggle to fully capture the subtle and complex features that characterize reversal patterns, particularly those that arise from the interplay of investor psychology and market dynamics. Human traders often rely on visual pattern recognition to identify these patterns, drawing upon their experience and intuition to interpret chart formations. Simulating this visual pattern recognition ability using machine learning techniques offers a potentially more robust and adaptable approach to technical analysis.

To achieve this, we leverage the capabilities of Convolutional Neural Networks (CNNs), trained on historical price data, to detect reversal patterns directly from 2D price images. This image-based approach allows us to move beyond the limitations of rigid rule-based methods, capturing more nuanced features and potentially identifying signals that might be missed by traditional techniques. Furthermore, recognizing the critical role that news and sentiment play in shaping investor behavior and influencing market trends, we incorporate news sentiment analysis to augment signals derived from image recognition.

Recent studies, notably Jiang et al. (2023), have applied CNNs to directly predict stock price trends from candlestick charts. However, we contend that this direct approach, mapping visual inputs to future returns, is inherently limited. Although CNNs are an effective method for image classification, noisy images are destructive factors in the training of CNNs and significantly decrease the classification performance (e.g., Goodfellow et al., 2014). The notoriously high degree of noise in financial data makes it challenging to discern true signals from random fluctuations, often leading to overfitting and poor generalizability when attempting direct prediction.

Rather than directly predicting returns, we employ CNNs to detect patterns in the paths of stock returns, identifying recurring structures and shapes that are more stable and generalizable across different markets. This pattern detection approach filters out some of the noise, enabling more robust and interpretable analysis. Furthermore, our study uniquely integrates the analysis

of news sentiment, recognizing the significant influence of investor sentiment on market dynamics (Barber and Odean, 2008, Da et al., 2011). We hypothesize that combining visual pattern recognition with the emotional undercurrents captured by sentiment analysis can provide a more comprehensive and accurate view of market movements. Indeed, we find that news sentiment can reinforce the performance of technical patterns, with negative sentiment exerting a particularly strong influence. This result resonates well with the psychology literature (Baumeister et al., 2001) that negative news tends to evoke stronger emotional responses.

Through empirical analysis in the US and Chinese markets, we show that our CNN-based pattern detection model achieves high accuracy and often identifies signals earlier than traditional rule-based methods. By examining the effectiveness of detected patterns, we find patterns better predict returns for firms with high momentum, institutional ownership, and prior patterns in US, while in China, they are more effective for smaller firms with higher momentum and institutional ownership. Importantly, news sentiment significantly reinforces the performance of these patterns. This combined approach, which integrates technical analysis with sentiment, enables the construction of a profitable long-short portfolio strategy that outperforms traditional benchmarks.

Our contributions are based on a recent body of literature that examines the effectiveness of technical patterns and the use of deep learning in finance. The early work of Lo et al. (2000) introduced a systematic pattern recognition algorithm, which demonstrated the practical value of technical analysis. Subsequent studies, such as Savin et al. (2007), explored the profitability of specific patterns such as 'Head and Shoulders', often finding mixed results for their standalone predictive power but observing potential gains when integrated with broader market strategies. Technical patterns have also been shown to be effective in various asset classes, including foreign exchange markets (Neely et al., 1997, Schulmeister, 2009). Our work also contributes to the existing literature exploring the role of firm characteristics in the effectiveness of technical trading (Jegadeesh et al., 2004, Fong and Yong, 2005, Marshall et al., 2009, Qian et al., 2018).

More recently, advances in deep learning, particularly the development of CNNs, have spurred a surge in research exploring their applications in financial market prediction (Kim and Kim, 2019, Lee et al., 2019, Hoseinzade and Haratizadeh, 2019, Hu et al., 2018). Researchers have successfully employed CNNs to analyze time series data transformed into images, demon-

strating improved prediction accuracy (Chen et al., 2016, Ahlawat, 2016). Several studies have focused on predicting stock price changes using graph- or image-centric CNNs, often achieving superior performance compared to traditional machine learning techniques (Eapen et al., 2019, Tsantekidis et al., 2017, Sim et al., 2019, Gudelek et al., 2017, Sezer and Ozbayoglu, 2018, Chen et al., 2021, Steinbacher, 2023). However, as previously discussed, many of these studies rely on direct prediction models that map visual patterns to forecasts of future returns; a strategy we argue can be prone to overfitting and limited generalizability.

Our study addresses this limitation by focusing on pattern detection rather than direct prediction, aligning more closely with the principles of technical analysis, which emphasize the identification of recurring price patterns as indicators of potential market changes. In addition, we incorporate news sentiment analysis, recognizing the crucial role of investor psychology in shaping market behavior. This combined approach, which takes advantage of CNNs for visual pattern recognition and combines sentiment analysis insights, offers a novel and potentially more robust framework to understand and exploit technical patterns in financial markets.

The remainder of this paper is structured as follows. Section 2 details the definition of technical patterns and the architecture of our image recognition model. Section A presents our data set that covers both the US and Chinese markets. Section 3 provides an empirical analysis of our method. Section 4 concludes.

2 Methodology

We first introduce the algorithm to define and identify technical patterns with time series. Furthermore, we present the architecture of our image recognition model and describe the process of representing historical market data as an image input for the model.

2.1 Technical Patterns

Following the findings of Lo et al. (2000), our analysis includes four pairs of technical patterns that are among the most popular patterns of traditional technical analysis: double tops and bottoms; head and shoulders, inverted head and shoulders; broadening tops and bottoms; triangle tops and bottoms. Unlike the kernel smoothing technique proposed by Lo et al. (2000),

we employ Bollinger bands as a technical trading instrument to algorithmically define local extrema and identify these patterns. We first describe the local maxima and minima of the time series of the stock price obtained with the Bollinger bands (Bollinger, 1992). The upper bands (UB) and lower bands (LB) are defined as:

$$UB = \bar{X}_{t,t+N} + c\sigma_{t,t+N}, \tag{1}$$

$$LB = \bar{X}_{t,t+N} - c\sigma_{t,t+N}, \tag{2}$$

with $\bar{X}_{t,t+N} = \frac{1}{N}\sum_{i=1}^{N} X_{t+i}, \sigma_{t,t+N} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_{t+i} - \bar{X}_{t,t+N})}$, where X is the mid-price (the average of high price, low price and closing price) and the coefficient c set as 1.1 in our rule based on parameter tuning. If the price hits the upper band, the local minimum is the minimum price between the previous local maximum and this hitting point. Similarly, the local maximum is recognized with the lower band. Following Lo et al. (2000) with modifications, Figure 1 shows examples of the eight technical patterns. When plotting stock price patterns, there are also various other shapes emerging beyond the eight specific patterns. For these patterns, we introduce a new class, named "general," which we use to categorize unidentified plots during the training of the image recognition model. Consequently, if an image cannot be recognized as one of the eight targeted technical patterns, it will be classified as a general category.

2.2 Candlestick Plot

With the technical patterns identified by the rules, we generate image inputs for the recognition model, adhering to the examples presented in Figures 1. In particular, our main price plot uses candlestick charts. The high and low prices correspond to the top and bottom of the central vertical bar. In contrast, the open and close prices are represented by the upper and lower bounds of the wide body. Each input image encapsulates consecutive daily price candlestick bars of a fully recognized technical pattern. The dimensions of the images are standardized, while the time intervals in the images are based on the length of complete patterns. Given that the candlestick plot inherently encapsulates pattern and trend information, we employ black-and-white imagery instead of green and red. This allows us to focus on two-dimensional pixel matrices rather than having to track a third dimension for RGB pixel intensities (same idea as Jiang et al., 2023), which can mitigate the complexity of the model.



(a) Double Tops and Double Bottoms



(b) Head And Shoulders and Inverted Head And Shoulders



(c) Broadening Tops and Broadening Bottoms



(d) Triangle Tops and Triangle Bottoms

Figure 1. Examples of technical patterns considered in our analysis.

2.3 The Image Recognition Model

We chose CNN as our image recognition model (Goodfellow et al., 2016). CNNs represent a regularized variant of multilayer perceptrons. They employ convolution operations instead of fully connected layers to process input data, thereby focusing on localized regions rather than the entire image. For example, considering images with fixed dimensions of 250×150 pixels (with a depth of 3 for RGB channels or 1 for grayscale images), there would be $250 \times 150 \times 3$ weight connections required to link the input layer to a single neuron. This requires the estimation of $250 \times 150 \times 3$ weight parameters. In the context of image classification tasks, a sufficient number of neurons is essential for the model to effectively learn features, consequently leading to an exponential increase in the number of weight parameters. Therefore, convolutional operations provide a more computationally efficient approach to addressing such challenges.

The CNN architecture includes an input layer, multiple hidden layers, and an output layer. The hidden layers comprise convolutional layers, pooling layers, and fully connected layers. Figure 2 shows the general comprehensive structure for image classification. The input and output layers represent the input data (image pixels) and the output predictions (labels). The components of the hidden layers represent the fundamental operations within a CNN, and we provide a detailed introduction in the following sections.

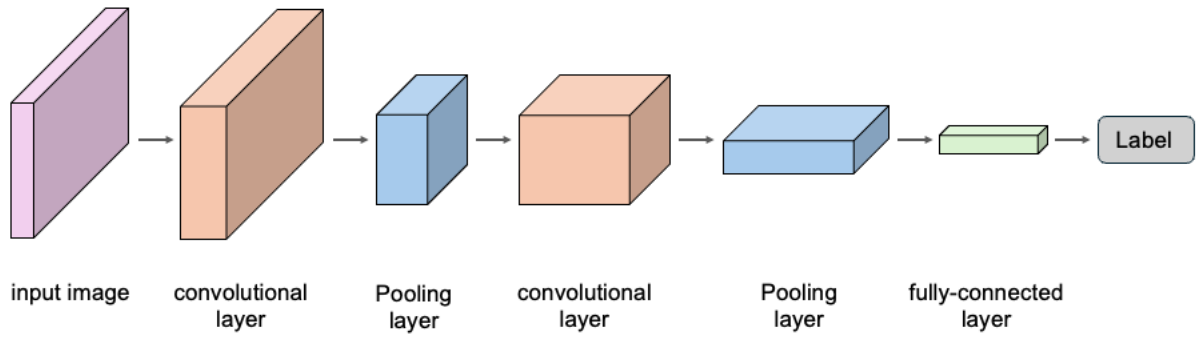


Figure 2. CNN Structure. An example for general CNN structure

2.3.1 Convolutional Layer

As the nomenclature suggests, the convolutional layer constitutes the principal operational component of a CNN. It employs kernels to perform convolution operations on the pixel matrix, subsequently applying nonlinear activation functions to the resultant matrix to ascertain the output of the neurons. The kernel, invariably a small matrix, traverses the entire input of the fixed neuron to compute the convolution. At this juncture, additional hyperparameters, namely stride and padding, are utilized to optimize the CNN by reducing the number of parameters or mitigating adverse effects. The stride parameter dictates the kernel's movement across the matrix, such that if the stride is denoted as s, the kernel advances by s elements per step. Concurrently, zero-padding is employed to preserve the information at the image borders (pixel matrix borders) by appending zeros beyond the original matrix boundaries. The zero-padding parameter p specifies the extent to which the image matrix size is augmented. Given an image size of $n \times n$ and a kernel size of $k \times k$, the output size is computed as follows:

$$O = 1 + \frac{n + 2p - f}{s}. \tag{3}$$

After the convolution operation, each kernel has a corresponding non-linear function (also called the activation function) to adjust the output. Common choices for activation functions are the Sigmoid, ReLU, and Tanh functions.

2.3.2 Pooling Layer

Pooling layers are intended to reduce the dimension of the input for subsequent layers and reduce the computational complexity of the model. It operates on the output following the activation map, incorporating max-pooling, L1/L2-normalization, and average pooling. However, the common choice in CNN is max-pooling, which returns the kernel's maximum value of the subregion. In the pooling layer, we also have stride and padding with effects similar to the convolution layers. If we use a pooling kernel of the size 2×2 with stride set to 2, then we can reduce the size of the output of this layer by 25% of its original size.

2.3.3 Fully Connected Layer

A fully connected layer in a CNN is a dense layer in which each neuron is connected to every neuron in the previous layer. This enables the network to integrate and process the features extracted by the earlier convolutional and pooling layers. This layer flattens the multidimensional input into a one-dimensional vector, performs a weighted sum of inputs, applies an activation function, and is crucial for making high-level predictions and classifications. It effectively translates spatially distributed features into a final decision, commonly used in the final stages of CNNs for tasks such as image recognition and classification. This is similar to the connection of neurons in a classic artificial neural network (ANN).

2.3.4 Parameters Choices

Next, we describe our model specification and training. The images in our dataset have a fixed size, a 250-pixel horizontal axis, and a 150-pixel vertical axis, represented as a matrix of normalized pixel values. We used black-and-white images to train the model because the shape of the candlestick plot has already revealed the pattern type without the color of the daily charts. So, the depth (channel) of the pixel matrix in our model is one, and the dimensions of our input data are $250 \times 150 \times 1$.

The kernel in our convolution layers is of size 5×5, and the kernel in our pooling is of size 2×2. The stride is set to 1 in the convolution layers and 2 in the pooling layers, and we utilize both the Sigmoid and the ReLU functions as the activation functions in different layers. The 2D matrix is transferred to the 1D vector with a flattening operation and half of the parameters are dropped in the fully connected layers. As discussed in Zeiler and Fergus (2014), the deeper layer learns more complex features, and the number of filters increases as the layer deepens. So, our model has 32, 64, and 1024 filters for the hidden layers. The architecture of our CNN model is shown in Figure 3 as described.

We divided the US stock data from 1992 to 1999 into training and testing samples, allocating 70% for training and 30% for testing without overlap. The samples in the test set are strictly from later dates than those in the training set. In the training step, we assign 10% of training samples as validation samples randomly. Due to the uneven distribution of rule-based technical

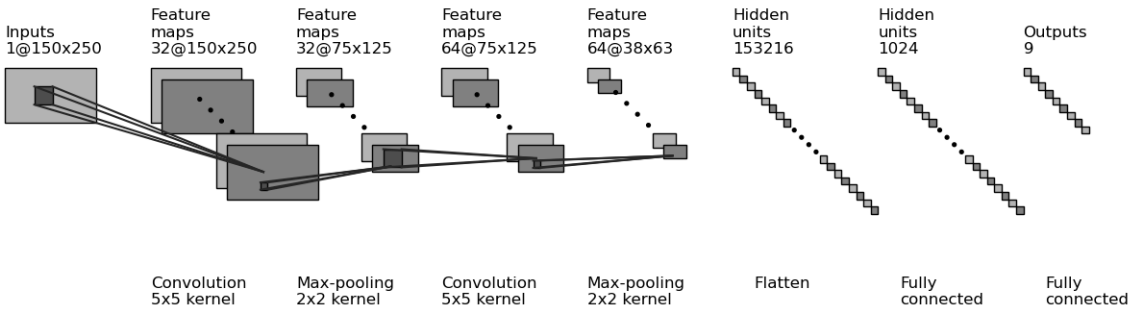


Figure 3. Structure of CNN model. This figure shows the structure of our CNN model. The input data is an image with a size of 150x250 and one dimension. In each feature map, the notation N@HxW means the number of filters is N with height H and width W in each filter. The output of the last CNN feature map is flattened to a vector and fed to a fully connected layer, where the reported input size (after the fully connected layer) is calculated as the product HxWxN from the last CNN building block (38x63x3). And the final output size is 9 because we have 9 classes.

patterns, we performed oversampling on the images in the training and validation sets to balance the number of each label in our classification problem. The resulting training and validation images have even numbers in every class. We train the model only once rather than retraining the model. The trained CNN model is then held fixed for the 2000 to 2021 US samples and the entire Chinese data. Since this is a multiclass classification problem, the model minimizes the categorical cross-entropy loss function in the training step. The loss function is defined as:

$$L(y, \hat{y}) = -\sum_{i=1}^{\#classes} y_i \ln \hat{y}_i, \tag{4}$$

where y is the true label and $\hat{y}$ is the output probability from the CNN model. If the image belongs to class k, then

$$y_k = 1, \tag{5}$$

$$y_i = 0, \ \forall i \neq k. \tag{6}$$

We use the Adam algorithm (Kingma and Ba, 2014) to minimize the loss function with a batch size of 128. The learning algorithm works three times through all training samples.

2.3.5 Model Visualization

This section presents a clear example of visualizing intermediate features within the CNN. This visualization facilitates a deeper understanding of the internal mechanisms by which the CNN classifies pattern images. Initially, we exhibit all the features that followed the first convolutional layer, as illustrated in Figure 4. Given that our CNN model incorporates 32 filters in the initial layer, there are consequently 32 feature map images for each pattern depicted in Figure 4.

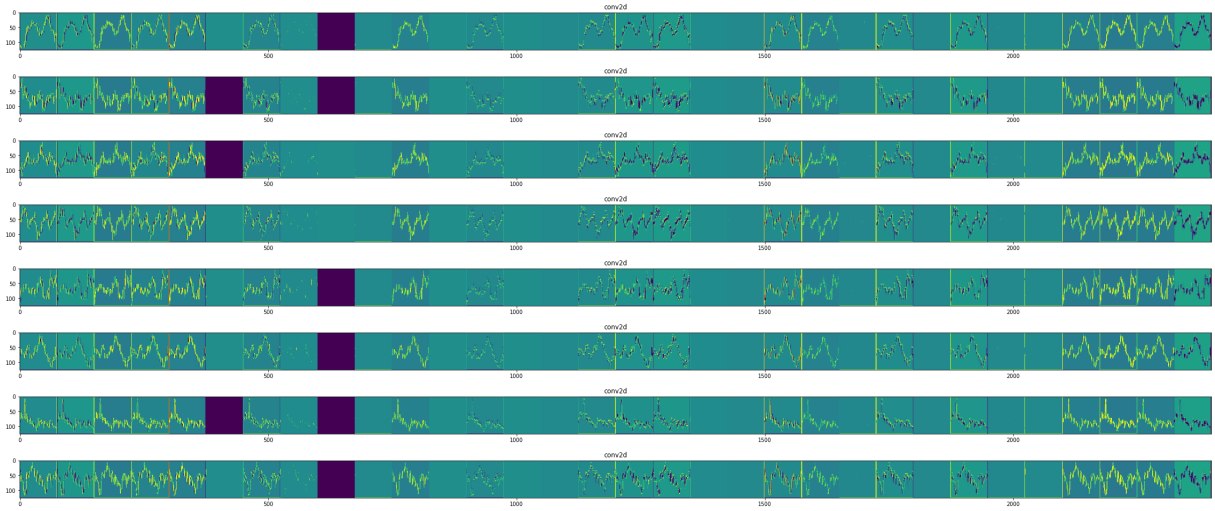


Figure 4. Data visualization examples. Feature maps after the first convolutional layer for the eight patterns. From top to bottom: double tops, double bottoms, head and shoulders, inverted head and shoulders, broadening tops, broadening bottoms, triangle tops, and triangle bottoms.

And for the features after the last convolutional layer, Figure 5 shows one feature for each pattern for clearer visualization. It illustrates that the local extrema and the overall shape are prominent in the feature maps to recognize a pattern without the impact of the noise.

2.3.6 Data Augmentation

As the rule-based algorithm can only identify a finite set of predefined patterns, we implement two data augmentation strategies for the training set to enhance the robustness of the CNN model. These augmentation techniques are applied to all samples from 1992 to 1999.

Taking advantage of the CNN's capacity to recognize invariant features from images irrespective of the degree of zoom, we first flatten images. Secondly, we augment the data by adding more daily prices before and after the recognized patterns as noise to the original plots

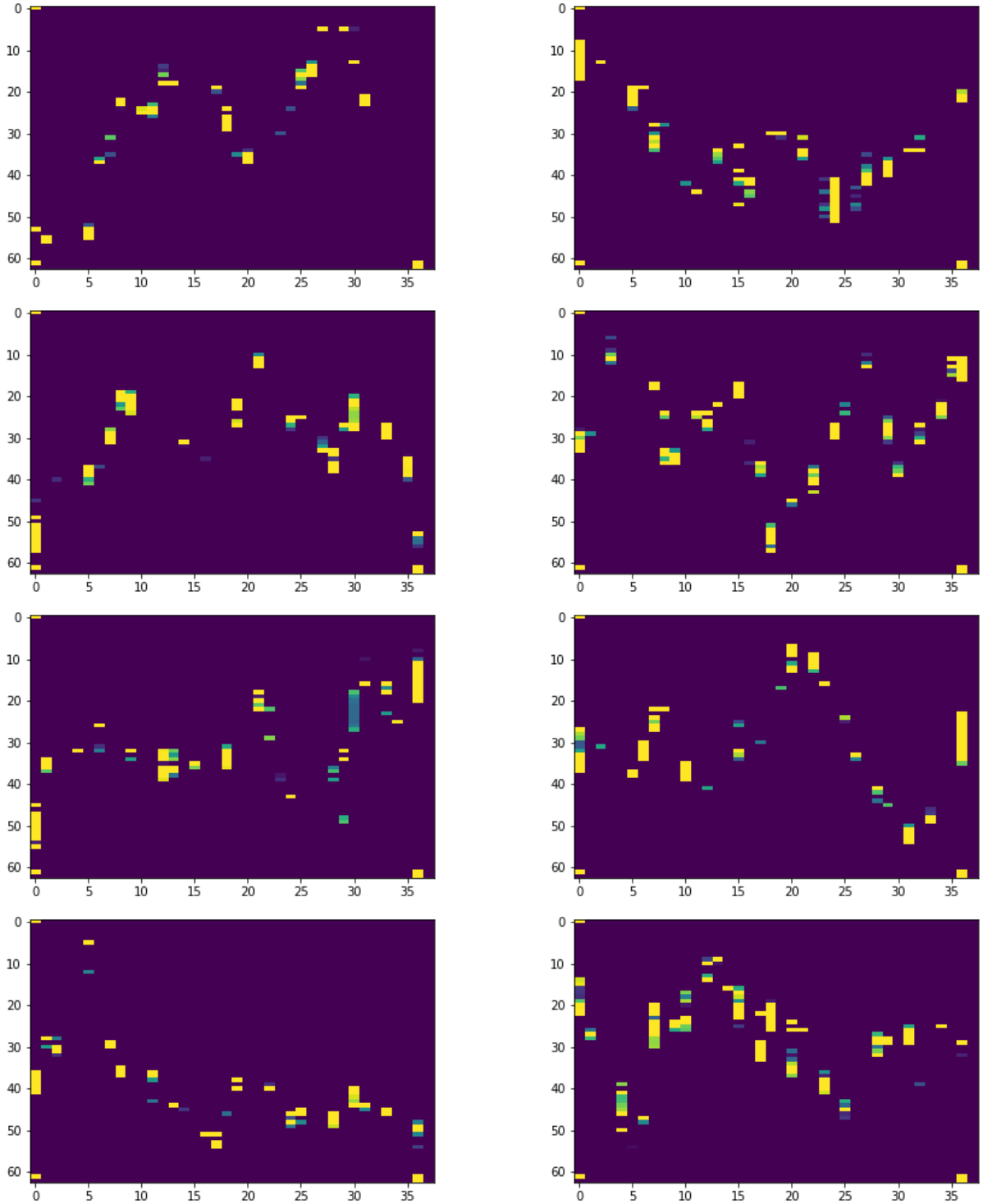


Figure 5. Data visualization examples. Feature maps after the last convolutional layer for the eight patterns. From top to bottom: double tops, double bottoms, head and shoulders, inverted head and shoulders, broadening tops, broadening bottoms, triangle tops, and triangle bottoms.



Figure 6. Data augmentation example. From left to right: the original pattern, the flattened pattern, the added pattern.

to improve the prediction ability of the CNN model. Figure 6 shows an example of data augmentation with these two approaches. These data augmentation techniques provide the model with a broader dataset for learning pattern features, preventing the model from overfitting.

2.4 News Sentiments

We briefly outline the methodology for constructing news sentiment signals in both the US and Chinese markets. By applying sentiment analysis to both markets, we aim to construct signals that capture the positive or negative news sentiments of firms. In the subsequent sections, we will delve into a more comprehensive analysis of the news' impact on technical patterns in the US and Chinese markets.

Notably, the news data are sparser in the US market, while it demonstrates greater density in the Chinese market. In both markets, we consider news that appears both before and after the pattern signals, encompassing a time frame of five days. If the news appears within five days before the pattern, we designate the pattern's end date as the signal date. In contrast, if the news comes within five days after the pattern, we define the news date as the signal date. When multiple news events occur within this period, priority is given to those preceding the recognized pattern date, with preference for the news closest to the pattern recognition date.

We establish specific criteria for validating event signals to ensure an effective combination of patterns and news sentiment. In particular, we identify tops, such as double tops, head and shoulders, broadening tops, and triangle tops, coupled with negative news as valid event signals. Similarly, we designate bottoms, such as double bottoms, inverted head and shoulders, broadening bottoms, and triangle bottoms, paired with positive news as valid event signals.

Using these different methodologies and criteria, we aim to effectively capture and analyze

the relationship between technical patterns and news sentiment in both the US and Chinese markets. This comprehensive approach allows us to draw valuable insights from the interplay between news sentiment and technical analysis, facilitating a deeper understanding of the dynamics within each market.

3 Empirical Analysis

We start our analysis by presenting the performance of our CNN model and comparing the patterns recognized by the CNN model (model-based results) with patterns identified by the time series algorithm (rule-based results) in terms of recognized quantity, coverage, and identification sequences.

3.1 Model Performance

We use the test set during 1992-1999 (disjoint with the training samples) to evaluate the performance of the model before applying the model to the entire sample. As shown in Figure 7, diagonal values dominate the confusion matrix, which means that most predicted patterns are the same as the true ones. The multi-class prediction accuracy based on the confusion matrix is defined as

$$Accuracy = \frac{\#\{predicted\ label = true\ label\}}{\#\{all\ samples\}}. \tag{7}$$

Our out-of-sample accuracy is 83.60%.

The model-based images are generated for all US common stocks from 2000 to 2021 and all China A-shares from 2005 to 2021, with daily rolling as the model input for further analysis.

3.2 Classification Comparison

We apply both rule-based and model-based approaches to identify technical patterns in US stock data over the entire 2000 to 2021 US sample and the entire 2005 to 2021 Chinese sample and subsequently compare the number of recognized patterns. In this analysis, we find that the CNN model not only recognizes patterns identified by rule-based methods but also detects additional patterns.

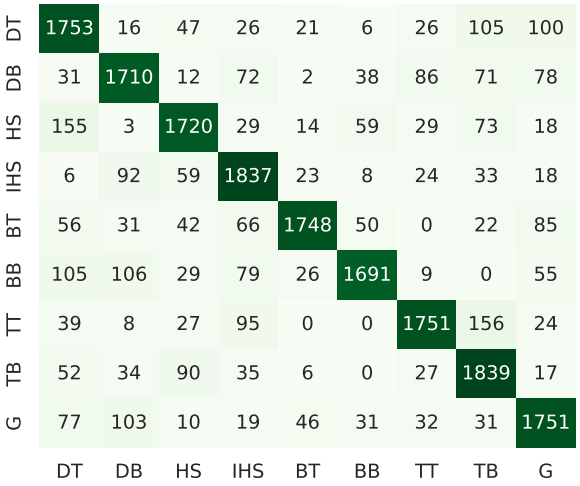


Figure 7. confusion matrix. From left to right, from top to bottom: double tops (DT), double bottoms (DB), head and shoulders (HS), inverted head and shoulds (IHS), broadening tops (BT), broadening bottoms (BB), triangle tops (TT), triangle bottoms (TB), general (G).

Our US dataset comprises 16,134,356 company image-end-date pairs, while our Chinese dataset includes 9,839,665 pairs as model input. In Tables 1 and 2, we report frequency counts for the number of patterns detected with different methodologies. As we combine technical patterns and news sentiment in our main analysis, we also present the number of combined signals in these two tables. We call the signal constructed with patterns and news sentiments pattern-sentiment signals. Note that news sentiments are available from 2003 in the US market and from 2014 in the Chinese market.

Pattern	**Rule**	**Model**	**Model + Sentiment**
Double Tops	7,705	88,952	9,708
Double Bottoms	6,204	72,695	13,071
Head and Shoulders	2,429	123,744	11,545
Inverted Head and Shoulders	2,233	134,402	27,736
Broadening Tops	2,880	29,509	2,450
Broadening Bottoms	2,532	67,250	12,296
Triangle Tops	2,623	39,980	3,763
Triangle Bottoms	2,732	31,236	7,313

Table 1. Number of recognized patterns with different methods and constructed signals, US market. This table reports the number of the rule-based patterns, the model-based patterns, and the pattern-sentiment combined signals. Note that the sample period for 'Model+Sentiment' signals starts in 2003.

As shown in Tables 1 and 2, the model identifies a significantly greater number of technical

Pattern	Rule	Model	Model + Sentiment
Double Tops	3,222	51,699	5,190
Double Bottoms	4,146	64,446	32,733
Head and Shoulders	823	88,803	8,829
Inverted Head and Shoulders	847	98,853	46,963
Broadening Tops	1,352	33,824	3,542
Broadening Bottoms	1,177	61,581	26,824
Triangle Tops	1,423	46,104	5,926
Triangle Bottoms	1,546	29,503	14,857

Table 2. Number of recognized patterns with different methods and constructed signals, Chinese market. This table reports the number of the rule-based patterns, the model-based patterns, and the pattern-sentiment combined signals. Note that the sample period for 'Model+Sentiment' signals starts in 2014.

patterns compared to the rule-based approach in both the US and Chinese markets, particularly for head and shoulders formations and inverted head and shoulders. This outcome suggests that the CNN model can de-noise and recognize potential patterns overlooked by the rule-based method, supporting our argument that CNN models with images are more suitable for recognizing technical patterns. Following the construction of new signals, the frequency of events for each pattern diminishes relative to the model-based results due to the sparsity of the news data in the US market and data availability in the Chinese market. However, the quantity of new signals remains generally higher than that of the rule-based patterns. Furthermore, in the Chinese news data, there are 270,999 positive sentiments while only 64,506 sentiments from 2014, leading to the observation in Table 2 that there are significantly more bottoms than tops after the constructed signals.

As indicated in Tables 1 and 2, the CNN model identifies significantly more patterns than the rule-based approach. Consequently, we examine the overlap between rule-based and model-based results. This analysis focuses on samples in the US market as an insight. For a given rule-based image, a matched image in the model-based results is defined as the same image ending date and the closest window length. We call them a matched pair. Given the window length choices of the images, the starting dates for the two patterns in a matched pair are likely to differ. Our findings reveal that 99.95% of rule-based results (29,323) have a corresponding model-based pattern.

There are two approaches to ascertaining patterns based on probability results: patterns with the highest probability or patterns with probabilities exceeding a specified threshold. Using

the first approach, 62.75% of matched pairs exhibit the same recognized patterns, whereas the second approach, with a threshold of 0.99, yields 81.53% matched pairs with identical patterns. Considering the discrepancies between the images in a matched pair and the accuracy of the CNN model, this outcome demonstrates that a substantial proportion of rule-based patterns can be identified by the model. For subsequent model-based results, patterns are determined using the first approach (patterns with the highest probability) for convenience. Among the pairs identified, the model recognizes 66.39% of the images previously identified by the model. This implies that the model consistently identifies the same patterns prior to the rule-based determination date. The earlier signals provided by the CNN model could be advantageous in trading strategies.

3.3 Cumulative Abnormal Returns

For the cumulative abnormal return (CAR) analysis, we use the capital asset pricing model (CAPM) as a benchmark to calculate abnormal returns. The corresponding CAR is defined as:

$$R_{i,t} = \mathbf{E}[R_{i,t}] + e_{i,t}, \tag{8}$$

$$\mathbf{E}[R_{i,t}] = R_{f,t} + \beta(R_{m,t} - R_{f,t}), \tag{9}$$

$$AR_t = \frac{1}{N}\sum_{i=1}^{N} e_{i,t}, \tag{10}$$

$$CAR(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_t, \tag{11}$$

where $R_{i,t}$ is the return of stock i, $R_{m,t}$ is the market return, $R_{f,t}$ is the risk-free rate, AR_t is the abnormal return at time t, and $CAR(t_1, t_2)$ is the cumulative abnormal return from day t_1 to t_2. To test the statistical significance of CAR, the t-test statistics is defined as

$$t_{CAR} = \frac{CAR(t_1, t_2)}{[\sigma^2(t_1, t_2)]^{\frac{1}{2}}}, \tag{12}$$

where $\sigma^2(t_1, t_2)$ is the variance of abnormal returns. The test statistics is typically assumed to follow a standard normal distribution in the absence of abnormal performance.

In our analysis, we consider each technical pattern as an independent event. Consequently, there are eight events in our analysis. We test the performance of short-term and long-term abnormal returns with 5, 10, 21, and 42-day holding periods following events. We first consider

each technical pattern as an independent event. To test our hypothesis that news sentiments can reinforce the performance after technical patterns, we consider events as tops (double tops, head and shoulder, broadening tops and triangle tops) with negative news sentiments and bottoms (double bottoms, inverted head and shoulder, broadening bottoms and triangle bottoms) with positive news sentiments.

3.3.1 US Market

We first report the performance of the short-term and long-term abnormal returns after both pattern-only and pattern-sentiment events in the US market.

Pattern	1-Day	5-Day	10-Day	21-Day	42-Day
Double Tops	-0.0006^{***}	-0.0001	-0.0004	0.0012^{**}	0.0038^{***}
	(-4.75)	(-0.47)	(-0.91)	(2.19)	(5.05)
Double Bottoms	0.0007^{***}	0.0018^{***}	0.0039^{***}	0.0077^{***}	0.0145^{***}
	(2.98)	(4.15)	(6.36)	(9.03)	(12.11)
Head and Shoulders	-0.0004^{***}	-0.0003	-0.0004	0.0020^{***}	0.0054^{***}
	(-3.60)	(-1.42)	(-1.13)	(4.05)	(7.62)
Inverted Head and Shoulders	-0.0000	0.0004^{*}	0.0016^{***}	0.0030^{***}	0.0069^{***}
	(-0.23)	(1.85)	(4.75)	(6.07)	(9.96)
Broadening Tops	0.0002	0.0009	0.0010	0.0002	0.0039^{***}
	(0.43)	(1.44)	(1.24)	(0.22)	(2.67)
Broadening Bottoms	0.0002	0.0009^{**}	0.0013^{**}	0.0032^{***}	0.0104^{***}
	(1.00)	(2.12)	(2.21)	(3.79)	(8.78)
Triangle Tops	-0.0009^{***}	-0.0004	-0.0003	0.0014	0.0099^{***}
	(-3.49)	(-0.70)	(-0.41)	(1.22)	(6.09)
Triangle Bottoms	0.0012^{***}	0.0025^{***}	0.0037^{***}	0.0068^{***}	0.0119^{***}
	(6.11)	(5.88)	(6.14)	(7.59)	(9.55)

Table 3. Cumulative abnormal returns (CAR), pattern-only events, US market. This table reports CARs and the statistical significance test statistics with 5, 10, 21, 42-day holding periods after eight pattern-only events. *: 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the CARs.

According to the definition of the eight technical patterns, stock abnormal returns after tops (double tops, head and shoulders, broadening tops, triangle tops) are expected to be negative while stock stock abnormal returns after bottoms (double bottoms, inverted head and shoulders, broadening bottoms, triangle bottoms) are expected to be positive if signals work effectively. Tables 3 and 4 reports the Cumulative Abnormal Returns (CAR) test statistics for both pattern-only and pattern-sentiment events in the US market, respectively. Note that we only consider model-based patterns in our main analysis. For the 0.05 significance level, if the absolute value of the statistic is higher than 1.96, we regard it as statistically significant. From the tables,

Pattern	1-Day	5-Day	10-Day	21-Day	42-Day
Double Tops	-0.0016^{***}	-0.0026^{***}	-0.0036^{***}	-0.0033^{**}	-0.0018
	(-3.63)	(-3.07)	(-3.32)	(-2.21)	(-0.88)
Double Bottoms	0.0011	0.0024^{**}	0.0034^{**}	0.0035^{**}	0.0057^{**}
	(1.44)	(2.17)	(2.55)	(2.06)	(2.51)
Head and Shoulders	-0.0020^{***}	-0.0022^{***}	-0.0015	0.0005	0.0024
	(-5.54)	(-3.10)	(-1.47)	(0.33)	(1.24)
Inverted Head and Shoulders	0.0009^{***}	0.0013^{***}	0.0022^{***}	0.0029^{***}	0.0036^{***}
	(4.24)	(3.16)	(4.01)	(3.49)	(3.15)
Broadening Tops	-0.0017^{**}	-0.0017	0.0005	0.0018	0.0023
	(-2.08)	(-1.08)	(0.24)	(0.55)	(0.52)
Broadening Bottoms	0.0004	0.0007	0.0004	0.0013	0.0076^{***}
	(1.25)	(0.98)	(0.45)	(0.89)	(3.53)
Triangle Tops	-0.0023^{**}	-0.0029	-0.0024	-0.0014	0.0033
	(-2.55)	(-1.41)	(-0.93)	(-0.42)	(0.71)
Triangle Bottoms	0.0008^{**}	0.0022^{***}	0.0032^{***}	0.0044^{***}	0.0056^{***}
	(2.10)	(2.97)	(3.17)	(3.11)	(2.79)

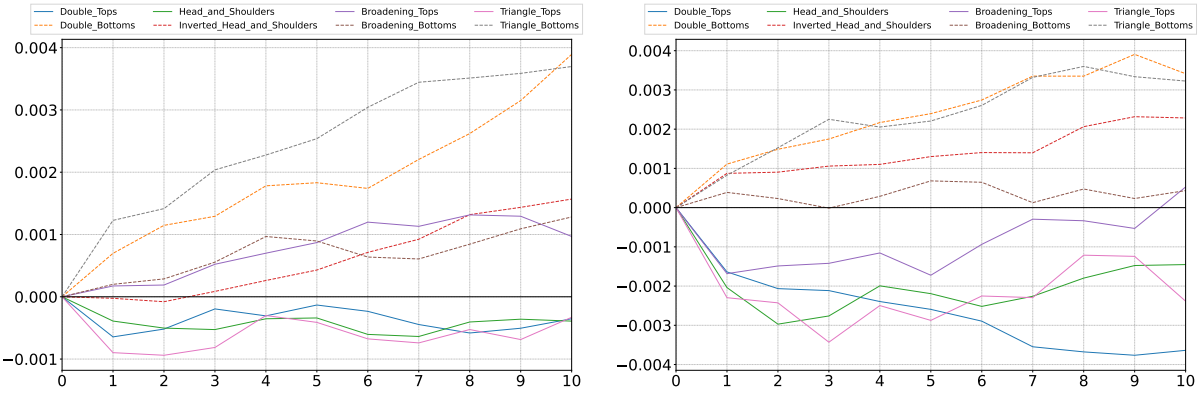
Table 4. Cumulative abnormal returns (CAR), pattern-sentiment events, US market. This table reports statistical significance test statistics of CARs with 5, 10, 21, 42-day holding periods after eight constructed pattern-sentiment events. *: 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the CARs.

we find that the significance of CARs varies among different patterns and different holding periods. In Table 3, test statistics for each pattern show reasonable signs in many days but lack significance in most of days. Patterns except for broadening tops have expected significant CARs on at least one day. This suggest that technical patterns with image recognition method work in many cases.

After involving news sentiments in Table 4, all tops (double tops, head and shoulders, broadening tops, triangle tops) show negative CARs while all bottoms (double bottoms, inverted head and shoulders, broadening bottoms, triangle bottoms) show positive CARs in 1 to 5 days as expected. However, in Table 3, signs of CARs broadening tops and bottoms are all positive within the 1 to 5 day period. Thus, the constructed pattern-sentiment signals demonstrate a more consistent performance across all four pattern pairs. In general, all tops and bottoms exhibit reasonable signs after pattern-sentiment events, with most showing significance within the first five days and a decay over longer holding periods. But the performance across different pattern pairs in Table 3 varies markedly, with notable differences observed in broadening patterns and other pairs. This supports our hypothesis that news sentiments have a reinforcing impact on technical analysis and can improve the performance consistency of technical patterns.

Figure 8 shows the evolution of cumulative returns from day 1 to day 10 after the events

(Figure 8a for pattern-only events and Figure 8b for pattern-sentiment events). We focus on CARs in the first 10 days as our observations from Tables 3 and 4 indicate that signals demonstrate their primary effectiveness within this period and decay very fast in the following days. Consistent with the results in Tables 3 and 4, the performance of tops and bottoms becomes notably symmetric with respect to zero on the x-axis compared with patterns-only signals. This further support our argument that news sentiment can improve patterns' performance in technical analysis since tops and bottoms should have opposite behavior in principle.



(a) Pattern-only signal

(b) Pattern-sentiment signal

Figure 8. Cumulative abnormal returns evolution, US market. This figure reports the 10-day daily cumulative abnormal returns (CAR) of pattern-only events (Panel (a)) and pattern-sentiment events (Panel (b)).

3.3.2 Chinese Market

We focus on both US and Chinese market in our analysis. Tables 5 and 6 show the test statistics of short-term and long-term cumulative abnormal returns with 5, 10, 21, 42-day holding periods after pattern-only and pattern-sentiment events in the Chinese market.

For the pattern-only results in Table 5, all patterns except double tops and triangle tops show positive CARs on the following day of event despite the different scales for tops and bottoms. For tops, only double tops and triangle tops show significant positive CARs as expected in the first 5 days. In general, pure pattern signals seem not work well in the Chinese market, particularly for tops. However, after considering news sentiments in technical analysis, Table 6 shows that all patterns have significant expected CARs on the first date after the event and these trends remain for 5 days for most of the patterns (except double bottom and broadening

Pattern	1-Day	5-Day	10-Day	21-Day	42-Day
Double Tops	-0.0011^{***}	0.0011^{***}	0.0056^{***}	0.0137^{***}	0.0261^{***}
	(-7.03)	(3.24)	(11.92)	(20.38)	(27.83)
Double Bottoms	0.0001	0.0000	0.0002	-0.0004	0.0034^{***}
	(0.95)	(0.21)	(0.81)	(-0.38)	(4.18)
Head and Shoulders	0.0003^{***}	0.0020^{***}	0.0034^{***}	0.0073^{***}	0.0136^{***}
	(3.14)	(8.44)	(10.35)	(15.26)	(20.22)
Inverted Head and Shoulders	0.0004^{***}	0.0018^{***}	0.0034^{***}	0.0046^{***}	0.0109^{***}
	(3.85)	(7.77)	(10.13)	(9.60)	(16.13)
Broadening Tops	0.0010^{***}	0.0040^{***}	0.0069^{***}	0.0151^{***}	0.0207^{***}
	(4.84)	(9.06)	(11.33)	(17.20)	(17.07)
Broadening Bottoms	0.0005^{***}	0.0006^{**}	0.0037^{***}	0.0092^{***}	0.0123^{***}
	(4.08)	(2.50)	(9.97)	(16.82)	(15.82)
Triangle Tops	-0.0008^{***}	-0.0006^{*}	-0.0008^{*}	0.0022^{***}	0.0100^{***}
	(-5.69)	(-1.90)	(-1.82)	(3.69)	(11.71)
Triangle Bottoms	0.0011^{***}	0.0033^{***}	0.0063^{***}	0.0094^{***}	0.0166^{***}
	(4.63)	(6.41)	(9.16)	(9.66)	(12.39)

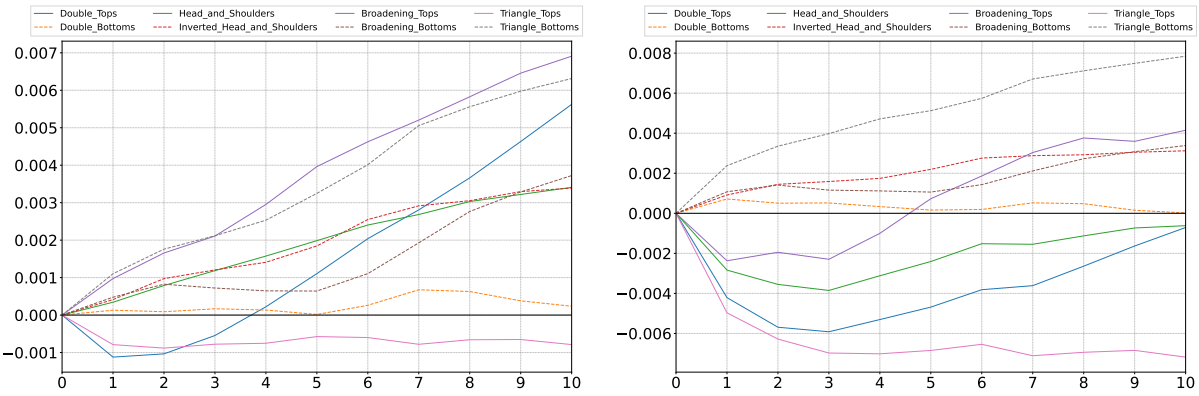
Table 5. Cumulative abnormal returns (CAR), pattern-only events, Chinese market. This table reports statistical significance test statistics of CARs with 5, 10, 21, 42-day holding periods after eight pattern-only events. *: 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the CARs.

Pattern	1-Day	5-Day	10-Day	21-Day	42-Day
Double Tops	-0.0042^{***}	-0.0047^{***}	-0.0007	0.0064^{***}	0.0182^{***}
	(-9.85)	(-5.30)	(-0.58)	(3.66)	(7.53)
Double Bottoms	0.0007^{***}	0.0002	0.0000	-0.0008	0.0021^{**}
	(3.87)	(0.43)	(0.03)	(-1.08)	(2.07)
Head and Shoulders	-0.0028^{***}	-0.0024^{***}	-0.0006	0.0035^{***}	0.0115^{***}
	(-9.75)	(-3.75)	(-0.69)	(2.68)	(6.31)
Inverted Head and Shoulders	0.0009^{***}	0.0022^{***}	0.0031^{***}	0.0038^{***}	0.0092^{***}
	(6.51)	(7.27)	(7.45)	(6.34)	(11.11)
Broadening Tops	-0.0024^{***}	0.0007	0.0041^{**}	0.0114^{***}	0.0153^{***}
	(-4.39)	(0.62)	(2.50)	(4.74)	(4.63)
Broadening Bottoms	0.0011^{***}	0.0011^{***}	0.0034^{***}	0.0076^{***}	0.0105^{***}
	(6.94)	(3.13)	(7.08)	(10.96)	(10.78)
Triangle Tops	-0.0050^{***}	-0.0068^{***}	-0.0072^{***}	-0.0001	0.0125^{***}
	(-13.70)	(-9.09)	(-6.88)	(-0.04)	(5.70)
Triangle Bottoms	0.0024^{***}	0.0052^{***}	0.0078^{***}	0.0103^{***}	0.0145^{***}
	(8.09)	(8.41)	(9.41)	(8.82)	(9.06)

Table 6. Cumulative abnormal returns (CAR), pattern-sentiment events, Chinese market. This table reports statistical significance test statistics of CARs with 5, 10, 21, 42-day holding periods after eight constructed pattern-sentiment events. *: 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the CARs.

tops). The trends decay as the holding period becomes longer. In technical analysis, positive returns are expected after bottoms and negative returns after tops immediately after pattern recognition, indicating the significant positive and negative CARs on day 1 after the events. Additionally, each pair of tops and bottoms in patterns, for example double tops and bottoms,

typically exhibit symmetric shapes. Therefore, the returns following these pattern signals are expected to display a form of symmetry around zero on average before signal decaying. In other words, this means that if there is a significant price decline after the formation of a top, the subsequent recovery after a bottom within the same pair should ideally show a comparable magnitude on average. The results in Table 6 satisfy these expectations during days 1 to 5, whereas Table 5 does not. Therefore, we state that the constructed pattern-sentiment signals show reinforcing function and work more effectively than pattern-only signals in the Chinese market.



(a) Pattern-only signal

(b) Pattern-sentiment signal

Figure 9. Cumulative abnormal returns evolution, Chinese market. This figure reports the 10-day daily cumulative abnormal returns (CAR) of pattern-only events (Panel (a)) and pattern-sentiment events (Panel (b)).

Figure 9 shows the evolution of cumulative returns from day 1 to day 10 after events in the Chinese market (Figure 9a for pattern-only signals and Figure 9b for pattern-sentiment signals). Consistent with results in Tables 5 and 6, the performance of combined signals is better than the pattern-only signals, particularly for double tops, inverted head and shoulders, and broadening tops. The results in the Chinese market further indicate that the impact of news sentiments in the technical analysis is consistent between different markets.

3.4 Test for News Sentiments Impact

In this section, we conduct formal tests to examine the impact of news sentiments on the performance of technical patterns. To distinguish between the effects of positive and negative news and to better understand their individual impact, we categorize patterns into two groups:

tops and bottoms. We categorize "Tops" to include patterns such as double tops, head and shoulders, broadening tops, and triangle tops, associated with negative news sentiments. In contrast, the "Bottoms" comprise double bottoms, inverted head and shoulders, broadening bottoms, and triangle bottoms, combined with positive news sentiments.

We use t-tests to analyze the difference between Tops (Bottoms) without any news and Tops with negative sentiments (Bottoms with positive sentiments). We consider patterns without any news instead of all patterns to avoid a correlation problem in the t-test statistics. The two-sample t-test is defined as

$$t_{tops,i} = \frac{\hat{x}_{tops,negative,i} - \hat{x}_{tops,no\ news,i}}{\sqrt{\frac{s^2_{tops,negative,i}}{n_{tops,negative,i}} + \frac{s^2_{tops,no\ news,i}}{n_{tops,no\ news,i}}}}, \tag{13}$$

and

$$t_{bottoms,i} = \frac{\hat{x}_{bottoms,positive,i} - \hat{x}_{bottoms,no\ news,i}}{\sqrt{\frac{s^2_{bottoms,positive,i}}{n_{bottoms,positive,i}} + \frac{s^2_{bottoms,no\ news,i}}{n_{bottoms,no\ news,i}}}} \tag{14}$$

for negative news sentiments (Tops) and positive news sentiments (Bottoms) respectively. In the formula, i represents the number of days after the events on which we focus. Panels A and B in Table 7 show the t-test statistics for negative and positive news sentiment effects in both US and Chinese markets.

In Panels A and B in Table 7, we find that negative sentiments exhibit a significantly strengthening effect on Tops and positive sentiments show a significantly positive impact on Bottoms in both the US and Chinese market. The strengthening impact of negative news tends to last longer and has a stronger impact than positive sentiments at first. To compare the reinforcing difference between negative and positive news sentiments, we further examine it with another t-test. The t-test statistics is defined as

$$t_i = \frac{|\hat{x}_{tops,negative,i} - \hat{x}_{tops,no\ news,i}| - |\hat{x}_{bottoms,positive,i} - \hat{x}_{bottoms,no\ news,i}|}{\sqrt{\frac{s^2_{tops,negative,i}}{n_{tops,negative,i}} + \frac{s^2_{tops,no\ news,i}}{n_{tops,no\ news,i}} + \frac{s^2_{bottoms,positive,i}}{n_{bottoms,positive,i}} + \frac{s^2_{bottoms,no\ news,i}}{n_{bottoms,no\ news,i}}}}, \tag{15}$$

where i represents the number of days after events. The last row of panels A and B in Table 7 reports the t-test statistics to compare the reinforcing impact between negative and positive sentiments. Consistently, the results indicate that negative sentiments have a statistically higher reinforcement potency in the technical analysis compared to positive news within the initial 10

days. This result is supported by the emotional impact in psychology literature (Baumeister et al., 2001) that negative news tends to evoke stronger emotional responses compared to positive news. This emotional intensity can make a more lasting impression on individuals. People may think more about negative events, leading to prolonged effects on their mood and behavior.

Panel A: US market

Sentiments	1-Day	5-Day	10-Day	21-Day
Negative	−4.95	−4.32	−3.47	−3.52
Positive	2.76	1.49	−0.11	−2.72
Difference	2.03	2.57	2.70	1.15

Panel B: Chinese market

Sentiments	1-Day	5-Day	10-Day	21-Day
Negative	−14.05	−10.65	−8.11	−7.51
Positive	5.00	−1.85	−6.18	-8.56
Difference	4.91	3.23	1.44	−0.26

Table 7. T test statistics for the reinforcement level of news sentiments. Panel (A) and Panel (B) shows the impact of positive and negative news sentiment on the performance of technical patterns in US (Panel A) and Chinese (Panel B) markets. The "difference" rows reports the difference of reinforcement levels between negative and positive news sentiments. Note that we test absolute negative impact - absolute positive impact here.

Given that pattern-sentiment signals exhibit significantly superior performance compared to pattern-only signals in both US and Chinese stock markets, our empirical analysis will focus on the combined signals in the following analysis.

3.5 Technical Analysis Effectiveness

In the realm of financial trading, the effectiveness of technical trading rules is not uniform across all stocks. In this section, we consider whether technical trading rules are more effective when applied to stocks of certain characteristic. We incorporate size and momentum as explanatory variables, given that firms with distinct attributes may exhibit differential price reactions to these patterns. We also consider institutional ownership to explore whether retail investors focus more on technical patterns than institutional investors. Furthermore, stocks with a high frequency of identified patterns may exhibit divergent price behaviors. Therefore, we include a variable to quantify the frequency of the pattern in our regression model. Based on the statistical significance of the CAR test metrics presented in Tables 4 and 6, we utilize

the CAR with a 5-day holding period following pattern-only events as the dependent variable in the subsequent regression analysis. We use pattern-only signals rather than pattern-sentiment signals to more broadly assess the effectiveness of technical patterns. Hence, we perform the following regression:

$$y_i = \alpha + \beta_{Size} Size_i + \beta_{Mom} Mom_i + \beta_{InstOwn} InstOwn_i + \beta_{NumPreYear} NumPreYear_i + \epsilon_i, \tag{16}$$

where the dependent variable y is a dummy variable representing whether the 5-day CAR has the expected sign. An effective pattern should satisfy the criterion that tops are associated with negative CARs and bottoms with positive CARs in the short term following the signals. Thus, y is defined as

$$y = \begin{cases} 1, & \text{if } CAR(5)_{tops} < 0 \text{ or } CAR(5)_{bottoms} > 0. \\ 0, & \text{otherwise,} \end{cases} \tag{17}$$

indicating the effectiveness of patterns. Independent variables in our analysis are firm size, momentum (Mom), institutional ownership (InstOwn), and number of patterns in the previous year (NumPreYear).

With this regression, we examine all pattern effectiveness on the individual stocks in both US and Chinese markets. The results are shown in Table 8. In the US market, firms with higher momentum, greater institutional ownership, and more detected patterns in the previous year, exhibit more effective performance with technical patterns. However, firm size has an insignificant impact, consistent with previous study Marshall et al. (2009). In the Chinese market, technical pattern trading rules tend to show better effectiveness in smaller firms with higher momentum and institutional ownership. Number of patterns in the previous year have negative influence though the effect is not significant. These results also align with the existing literature. Qian et al. (2018) finds that firms with higher excess profits from technical trading have higher institutional ownership and the excess profits of the highest technical trading profit quantile portfolio disappear in the following year in the Chinese market. Consistently, momentum and institutional ownership significantly and positively explain the effectiveness of pattern performance in US and Chinese markets. Although the adjusted R^2 suggests that the

effectiveness of patterns is not largely explained in both markets, our explanatory variables provide valuable insights, indicating that firms with certain characteristics have more effective technical trading rules.

	(1)	(2)
Const	0.5681***	0.5563***
	(769.309)	(635.023)
Size	0.0008	−0.0041***
	(1.520)	(−4.740)
Momentum	0.0047***	0.0111***
	(6.142)	(11.571)
InstOwn	0.0072***	0.0020**
	(9.579)	(2.309)
NumPreYear	0.0022***	−0.0005
	(2.925)	(−0.444)
No. Observations	454,100	323,360
Adj. R^2(%)	0.0310	0.0447

Table 8. Firm characteristic regression: The dependent variable is a dummy variable, indicating whether the 5-day CAR is with the expected sign. (1): US market, (2): Chinese market. *: 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the estimated parameters.

3.6 Portfolio Performance

So far, our assessment of signal performance has been entirely based on statistical tests. Next, we analyze whether pattern-sentiment signals can be exploited in portfolio strategies. In portfolio analysis, we combine all pattern-sentiment signals into one portfolio in the US and Chinese markets. Considering the total number of signals generated by model-based patterns and the sentiment of the news in Tables 1 and 2, it is feasible to create portfolios based on them.

We consider signals lasting one day when building our portfolio. Due to the availability of news sentiment data, our portfolio starts in 2003 for the US market and in 2014 for the Chinese market. To avoid excessively high positions in single stocks, we set the maximum weight of each stock as 0.5. Suppose the portfolio value for each kind of signal based on the recognized patterns at day t is $P_{t,j}$, defined as

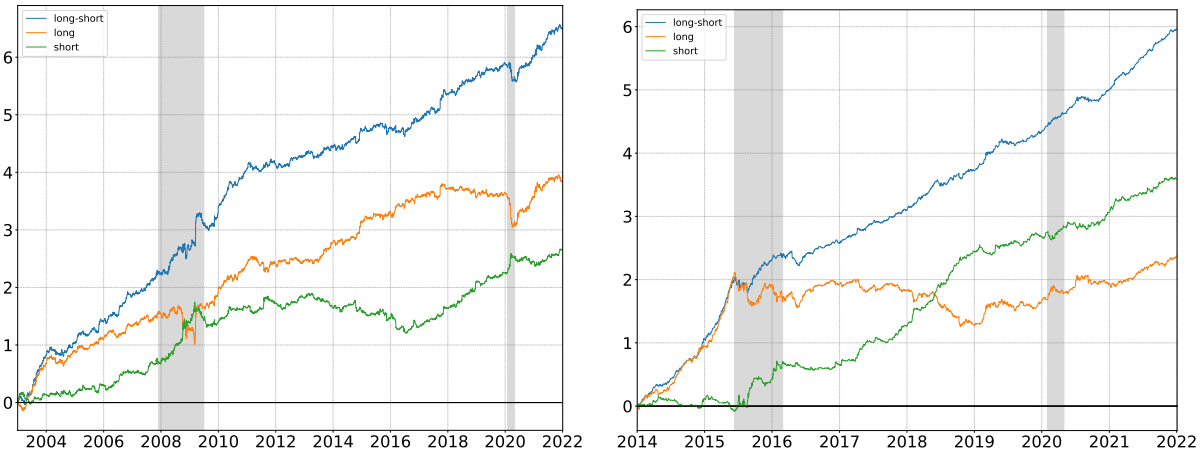
$$P_{t,j} = \sum_{i=1}^{n_{t,j}} w_{i,j} s_{i,j}, \quad w_{i,j} = \min\{1/n_{t,j}, 0.5\}, \tag{18}$$

where j represents the type of pattern-sentiment signal. $n_{t,j}$ is the number of this signal in day t and $s_{i,j}$ is the price of stock i identified as this signal.

Using these separate portfolios $P_{t,j}$ as a basis, we construct the final portfolio by combining all pattern-sentiment signals together. This composite portfolio involves buying stocks with signals of double bottoms, inverted head and shoulders, broadening bottoms, and triangle bottoms with positive sentiments, while shorting stocks with signals of double tops, head and shoulders, broadening tops, and triangle tops with negative sentiments. Therefore, the portfolio combining all signals is defined as:

$$OP_t = \frac{1}{4}\sum_{p\in\mathcal{P}} P_{p,t} - \frac{1}{4}\sum_{n\in\mathcal{N}} P_{n,t}, \tag{19}$$

where $\mathcal{P} = \{DB, IHS, BB, TB\}$ and $\mathcal{N} = \{DT, HS, BT, TT\}$ are the long and short signals respectively. We also control the recognized pattern duplication issue in our analysis.



(a) US market **(b)** Chinese market

Figure 10. Cumulative sum of portfolio return: the figure represents the cumulative sum of the portfolio returns combining all signals constructed with pattern and news sentiment signal in different markets. Panel (a): US market, Panel (b): Chinese market.

Panel (a) in Figure 10 illustrates the cumulative return for the portfolio constructed using all pattern-sentiment signals in the US market, displaying an upward trend over the long term. The cumulative sum of excess returns exceeds a value of 6 after holding this portfolio from 2003 to 2021, and generates the annualized Sharpe ratio of 1.38. Panel (b) in Figure 10 shows the cumulative sum of portfolio returns from 2014 to 2022 in the Chinese market. In line with the US market, the cumulative sum shows consistent growth over the years. In particular, the

annualized Sharpe ratio is 4.04, even exceeding the Sharpe ratio in the US market. This indicates a superior performance of our signals and strategy in the Chinese market and a consistent performance of our strategy across markets without retraining the model.

Next, we assess whether the returns of the pattern portfolio can be explained by exposure to well-known stock factors. Initially, we examine the Fama-French three-factor regression (Fama and French, 1993) in both markets. The three factors are returns of market portfolio (MKT), the historic excess returns of small caps over big caps (SMB) and of value stocks over growth stocks (HML). The regression is defined as

$$OP_t = \alpha + \beta_{MKT} r_{t,MKT} + \beta_{SMB} r_{t,SMB} + \beta_{HML} r_{t,HML} + \epsilon_t. \tag{20}$$

Given the results from Fama-French 3 factor model, we further consider alphas based on the Carhart 4-factor model, Fama-French 3 factors with the monthly premium on winners minus losers (UMD) (Carhart, 1997), to examine whether the portfolio returns can be explained by the momentum strategy. The regression is defined as

$$OP_t = \alpha + \beta_{MKT} r_{t,MKT} + \beta_{SMB} r_{t,SMB} + \beta_{HML} r_{t,HML} + \beta_{UMD} r_{t,UMD} + \epsilon_t. \tag{21}$$

	(1)	(2)
$\alpha(\%)$	0.1350^{***}	0.1353^{***}
	(6.064)	(6.086)
β_{MKT}	0.0431^{**}	0.0355^{*}
	(2.201)	(1.805)
β_{SMB}	0.1890^{***}	0.1830^{***}
	(4.811)	(4.921)
β_{HML}	0.0339	-0.0353
	(1.097)	(-0.950)
β_{UMD}		-0.0918^{***}
		(-3.361)
Adj. $R^2(\%)$	0.6439	0.8623

Table 9. Alphas, Betas and adjusted R^2s of the regressions of the portfolios on the classic models, US market. (1): Fama-French 3 factors, (2): Carhart 4 factors. *: 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the estimated parameters.

The Fama-French 3-factor and Carhart 4-factor regressions' alphas, betas, and adjusted R^2s in the US market are shown in Table 9. Alphas in 3-factor and 4-factor models both show significance with positive value, which means that our portfolio's returns cannot be entirely explained by the the classic model factors. In other words, the alphas represent the excess re-

turns of the portfolio over the explanation by the Fama-French model and Carhart model with exposure to market risk, size, value, and momentum factors and show the portfolio has outperformed expectations. Furthermore, the betas of the market return and size exhibit significant positivity in both cases. After adding momentum (UMD) to the regression, the betas of market return and size remain the same, while there is a slight change in the beta of the value factor. The momentum betas are statistically significant for all patterns. However, the adjusted-R^2 values are notably low in these regressions. This suggests that while the well-known factors can account for some of the excess returns, our pattern-sentiment signals provide additional information beyond that.

	(1)	(2)
$\alpha(\%)$	0.2849^{***}	0.2851^{***}
	(10.177)	(10.177)
β_{MKT}	0.2485^{***}	0.2484^{***}
	(12.547)	(12.539)
β_{SMB}	0.1149^{***}	0.1144^{***}
	(2.910)	(2.892)
β_{HML}	0.1657^{***}	0.1650^{***}
	(3.075)	(3.057)
β_{UMD}		-0.0009
		(-0.243)
Adj. $R^2(\%)$	8.6884	8.6400

Table 10. Alphas, Betas and adjusted R^2s of the regressions of the portfolios on the classic models, Chinese market. (1): Fama-French 3 factors, (2): Carhart 4 factors. 0.1 significance level, **: 0.05 significance level, ***: 0.01 significance level. The values in parentheses are the t-statistics of the estimated parameters.

In the Chinese market, Table 10 presents the results of the Fama-French 3-factor and Carhart 4-factor models. Consistent with the results of the US market, the alphas are significantly positive, indicating that our strategy also outperforms the Fama-French model on the Chinese market. The market return and size display significantly positive estimators, while the book-to-market ratio shows a significantly negative impact on the portfolio return. However, momentum does not provide any additional significant explanation. Thus, the strategy implemented with our pattern-sentiment signals show superior performance in both US and Chinese market. Furthermore, portfolio returns cannot be fully explained by the four classic factors.

4 Conclusion

We explore the validity of technical patterns in recent years and make a thorough inquiry on the impact of news sentiment on technical analysis in US and Chinese markets. Taking advantage of the image recognition method, we apply CNN with image input to recognize technical patterns. Instead of end-to-end prediction, we use CNN classifications as explainable features in further analysis. We show that the image recognition model has high accuracy in training samples and good performance in out-of-sample data with empirical analysis. In addition, the image recognition model can detect many more patterns and obtain earlier identification of many samples than the rule-based method with time series.

To evaluate the question whether news sentiments matter in technical analysis. We apply an event study with cumulative abnormal returns to compare and test the impact of news sentiments. We find that many model-based patterns show significant CARs with different holding periods in the US market, whereas top-type patterns seem to not work well in the Chinese market. However, the news sentiment can reinforce the performance of technical patterns in both markets. For pattern-sentiment events, in the US market, all tops exhibit negative CARs, while all bottoms display positive CARs in 1 to 5 days as expected. These results are statistically significant, except for the broadening bottoms with positive sentiment. In the Chinese market, all tops show significantly negative CARs and all bottoms show significantly positive CARs on the first day after signals. In general, the performance is as expected. Based on further tests, we find that news sentiment can reinforce the performance of technical patterns, with negative sentiment exerting a particularly strong influence.

Our study on portfolio performance, leveraging pattern-sentiment indicators, has shown improved outcomes in event studies. We infer that the combination of technical patterns with news sentiment continues to be effective and profitable in the modern stock market, surpassing the Fama-French factors. We employ straightforward images without incorporating moving averages or volume bars. However, the results of our models and empirical analysis demonstrate the predictive capabilities and profitability of technical pattern-based image recognition methods. Our findings suggest that the integration of image recognition techniques and sentiment is crucial in technical analysis.

A Data

A.1 Stock Imaging Data

We use daily stock data from CRSP for all firms for the US market. Our sample covers all common shares from June 1992 to December 2021 based on the fact that daily opening, high, and low prices become available from June 1992. The prices have been rigorously adjusted in accordance with the relevant adjustment factors and dividends information provided by CRSP. Moreover, our CNN model is trained on US stock price data spanning the period from 1992 to 1999.

In the Chinese market, we utilize daily stock data from JQData for all A-shares from January 2005 to December 2021. Our sample starts from 2005 due to the availability of price data in JQData. Our analysis is confined to A-share data for model application purposes without retraining another model for the Chinese market, examining of the consistency of our strategy.

A.2 News Sentiment

For US firms, we incorporate news data obtained from Refinitiv, which offers comprehensive news classifications (positive, negative, or neutral) for each news item. This dataset covers data beginning in January 2003. We calculate the news sentiment based on the frequency of positive and negative news items on each date t for every stock i. By aggregating all news articles for each company, consistent with prior studies (Yang et al., 2015), the news sentiment is defined as follows:

$$sent_{i,t} = \frac{\#\text{PositiveNews}_{i,t} - \#\text{NegativeNews}_{i,t}}{\#\text{TotalNews}_{i,t}}. \tag{22}$$

For the Chinese market, we collect news data from Tonglian, which provides sentiments already in their raw data. The news sentiments for most companies in the Chinese market are available from 2014, thus we conduct the empirical analysis for the Chinese market from 2014, integrating technical patterns and news sentiments in our main analysis.

A.3 Firm Characteristics

For firm characteristic factors, we include firm size, momentum (Mom), and institutional ownership percentage (IOP) in our further analysis. For the US market, firm size data is collected from CRSP, calculated with share prices and number of shares outstanding (Banz, 1981). Momentum is determined using stock price data from CRSP, calculated as the nearest annualized stock close-to-close return minus the nearest monthly stock close-to-close return (Jegadeesh and Titman, 1993). Institutional ownership data is obtained from the Thomson-Reuters Institutional Holdings (13F) Database. In the Chinese market, we collect firm characteristic data (size, Mom and IOP) from CSMAR. In Table 11, we provide summary statistics for the firm characteristics data. In the main analysis, we use normalized variables to ensure that variable scales are consistent in the regression.

Panel (A): US market

	No.	**Mean**	**Std**	**Min**	**25%**	**50%**	**75%**	**Max**
Size ($\times 10^8$)	79,731	102.23	464.83	0.00	2.44	11.24	48.78	26,340.47
Mom	86,505	0.11	0.92	−9.00	−0.19	0.03	0.26	116.39
InstOwn	56,539	0.71	6.42	0.00	0.52	0.74	0.87	1,079.03

Panel (B): Chinese market

	No.	**Mean**	**Std**	**Min**	**25%**	**50%**	**75%**	**Max**
Size ($\times 10^8$)	137,420	2.52	6.12	0.00	0.37	0.91	2.38	316.45
Mom	131,388	0.09	0.59	−2.43	−0.24	−0.04	0.25	17.49
InstOwn	144,019	0.44	0.25	0.00	0.23	0.45	0.63	1.00

Table 11. Descriptive statistics for firm characteristics data. Panel (A) reports the summary statistics for the US market and Panel (B) reports the summary statistics for the Chinese market.[1]

[1]These numbers are derived from the regression data for the pattern-sentiment signals in our main analysis. Although the US market data spans a longer time period, the number of observations is smaller. This is due to the sparser news data in the US market, resulting in fewer pattern-sentiment signals.

B Definition of Technical Patterns

The eight technical patterns in our paper are defined as following based on Lo et al. (2000):

Definition 1 (Double Tops, Double Bottoms)**.** Suppose E_1 is an initial local extrema and E_a, E_b are the subsequent local maximum and local minimum. t_1, t_a, t_b are the corresponding date and τ_a, τ_b are the pattern stopping times.

$$DT = \begin{cases} E_1 \text{ is a maximum}, \\ E_1,\ E_a \in [\bar{E}_{1,a} - 1.5\%\bar{E}_{1,a},\ \bar{E}_{1,a} + 1.5\%\bar{E}_{1,a}],\ \bar{E}_{1,a} = (E_1 + E_a)/2, \\ t_a - t_1 > 22, \\ \tau_a = \inf\limits_{t>t_a}\{p_t \text{ hits the neck line}\}, \end{cases}$$

$$DB = \begin{cases} E_1 \text{ is a minimum}, \\ E_1,\ E_b \in [\bar{E}_{1,b} - 1.5\%\bar{E}_{1,b},\ \bar{E}_{1,b} + 1.5\%\bar{E}_{1,b}],\ \bar{E}_{1,b} = (E_1 + E_b)/2, \\ t_b - t_1 > 22, \\ \tau_b = \inf\limits_{t>t_b}\{p_t \text{ hits the neck line}\}. \end{cases}$$

Definition 2 (Head and Shoulders, Inverted Head and Shoulders)**.** Suppose $\{E_i\}_{i=1}^5$ are five consecutive local extrema, then

$$HS = \begin{cases} E_1 \text{ is a maximum}, \\ E_3 > E_1,\ E_3 > E_5, \\ E_1,\ E_5 \in [\bar{E}_{1,5} - 1.5\%\bar{E}_{1,5},\ \bar{E}_{1,5} + 1.5\%\bar{E}_{1,5}],\ \bar{E}_{1,5} = (E_1 + E_5)/2, \\ E_2,\ E_4 \in [\bar{E}_{2,4} - 1.5\%\bar{E}_{2,4},\ \bar{E}_{2,4} + 1.5\%\bar{E}_{2,4}],\ \bar{E}_{2,4} = (E_2 + E_4)/2, \\ E_3 > (\bar{E}_{1,5} + 1.5\%\bar{E}_{1,5}), \end{cases}$$

$$IHS = \begin{cases} E_1 \text{ is a minimum}, \\ E_3 < E_1,\ E_3 < E_5, \\ E_1,\ E_5 \in [\bar{E}_{1,5} - 1.5\%\bar{E}_{1,5},\ \bar{E}_{1,5} + 1.5\%\bar{E}_{1,5}],\ \bar{E}_{1,5} = (E_1 + E_5)/2, \\ E_2,\ E_4 \in [\bar{E}_{2,4} - 1.5\%\bar{E}_{2,4},\ \bar{E}_{2,4} + 1.5\%\bar{E}_{2,4}],\ \bar{E}_{2,4} = (E_2 + E_4)/2, \\ E_3 < (\bar{E}_{1,5} - 1.5\%\bar{E}_{1,5}). \end{cases}$$

Definition 3 (Broadening Tops, Broadening Bottoms)**.** Suppose $\{E_i\}_{i=1}^5$ are five consecutive

local extrema, then

$$BT = \begin{cases} E_1 \ is \ a \ maximum, \\ E_1 < E_3 < E_5, \\ E_2 > E_4, \\ E_2 \geq \bar{E}_{2,4} - 1\%\bar{E}_{2,4}, \ E_4 \leq \bar{E}_{2,4} + 1\%\bar{E}_{2,4}, \ \bar{E}_{2,4} = (E_2 + E_4)/2, \end{cases}$$

$$BB = \begin{cases} E_1 \ is \ a \ minimum, \\ E_1 > E_3 > E_5, \\ E_2 < E_4, \\ E_2 \leq \bar{E}_{2,4} - 1\%\bar{E}_{2,4}, \ E_4 \geq \bar{E}_{2,4} + 1\%\bar{E}_{2,4}, \ \bar{E}_{2,4} = (E_2 + E_4)/2. \end{cases}$$

Definition 4 (Triangle Tops, Triangle Bottoms)**.** Suppose $\{E_i\}_{i=1}^5$ are five consecutive local extrema, then

$$TT = \begin{cases} E_1 \ is \ a \ maximum, \\ E_1 > E_3 > E_5, \\ E_2 < E_4, \\ E_2 \leq \bar{E}_{2,4} - 1\%\bar{E}_{2,4}, \ E_4 \geq \bar{E}_{2,4} + 1\%\bar{E}_{2,4}, \ \bar{E}_{2,4} = (E_2 + E_4)/2, \end{cases}$$

$$TB = \begin{cases} E_1 \ is \ a \ minimum, \\ E_1 < E_3 < E_5, \\ E_2 > E_4, \\ E_2 \geq \bar{E}_{2,4} - 1\%\bar{E}_{2,4}, \ E_4 \leq \bar{E}_{2,4} + 1\%\bar{E}_{2,4}, \ \bar{E}_{2,4} = (E_2 + E_4)/2. \end{cases}$$

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